

VIT BHOPAL UNIVERSITY

School of Computer Science & Engineering

VITyarthi E-Learning Platform

PROJECT REPORT

Resume Analyzer + Skill Gap Detector

CSA2001 – Bring Your Own Project (BYOP)

Field	Details
Student Name	Ankit Kumar Mandal
Registration No.	25BAI10217
Branch	CSE (AI & ML)
Year	First Year B.Tech
Course Code	CSA2001
Course Title	Bring Your Own Project (BYOP)
Project Title	Resume Analyzer + Skill Gap Detector
GitHub Repo	github.com/ankit-kr6/resume-analyzer
Tech Stack	Python Streamlit pdfplumber JSON
Submission Date	29th March 2026

1. ABSTRACT

The Resume Analyzer + Skill Gap Detector is a Python-based web application developed as part of the CSA2001 Bring Your Own Project (BYOP) assignment. The application addresses a common challenge faced by students and job seekers: identifying the gap between their current skillset and the requirements of their target job role.

Using Streamlit as the frontend framework and pdfplumber for PDF text extraction, the application provides an intuitive interface where users can upload their resume in PDF format, select a target job role, and instantly receive a detailed skill gap analysis including matched skills, missing skills, and a percentage match score with visual feedback. The project demonstrates practical application of file handling, string processing, data structures, and web application development concepts.

2. PROBLEM STATEMENT

In today's competitive job market, students and freshers often struggle to understand exactly which skills they need to develop to qualify for their target job roles. While job descriptions list required skills clearly, candidates frequently lack a systematic way to compare their existing skillset against these requirements.

Key problems identified:

- Students are unaware of specific skill gaps for their target roles
- Manual comparison of resume skills vs job requirements is time-consuming
- No easy way to quantify how suitable a resume is for a specific role
- Candidates waste time applying for roles they are not qualified for
- Lack of targeted guidance on which skills to learn next

3. OBJECTIVE

The primary objectives of this project are:

1. Build a fully functional web application that accepts PDF resumes as input
2. Extract and analyze text content from uploaded resumes using NLP techniques
3. Compare extracted skills against a curated job role skills database
4. Display matched skills, missing skills, and a quantified match score
5. Provide visual feedback to help users understand their profile strength

- 6. Ensure the application is fully executable via command line (streamlit run app.py)
- 7. Demonstrate practical application of Python, file handling, and web development

4. TECHNOLOGY STACK

Technology	Version	Purpose
Python	3.10+	Primary programming language
Streamlit	Latest	Web application framework - CLI executable
pdfplumber	Latest	PDF text extraction and parsing
JSON	Built-in	Job role to skills mapping database
Git & GitHub	Latest	Version control and repository hosting

5. SYSTEM ARCHITECTURE

Project Structure:

```
resume-analyzer/  
■■■ app.py # Main Streamlit application (entry point)  
■■■ analyzer.py # Core logic: PDF parsing + skill extraction  
■■■ skills_db.json # Database: job roles mapped to required skills  
■■■ requirements.txt # Python package dependencies  
■■■ sample_resume.pdf # Sample resume for evaluator testing  
■■■ README.md # Setup and run instructions
```

Data Flow Architecture:



6. FEATURES IMPLEMENTED

Feature	Description	Status
PDF Upload	Accepts any PDF resume file via browser interface	[DONE]
Text Extraction	Extracts all text from PDF using pdfplumber library	[DONE]
Skill Detection	Case-insensitive keyword matching for 30+ skills	[DONE]
6 Job Roles	Data Scientist, Web Dev, DevOps, Software Eng, Data Analyst, ML Eng	[DONE]
Match Score	Percentage score with animated progress bar	[DONE]
Visual Feedback	Green for matched skills, Red for missing skills	[DONE]
Score Feedback	Excellent/Good/Low messages based on score	[DONE]
CLI Executable	Single command: streamlit run app.py	[DONE]

7. SUPPORTED JOB ROLES & SKILLS DATABASE

Job Role	Required Skills
Data Scientist	Python, Machine Learning, SQL, Statistics, TensorFlow, Pandas, Data Visualization
Web Developer	HTML, CSS, JavaScript, React, Node.js, REST API, Git
DevOps Engineer	Docker, Kubernetes, CI/CD, Linux, AWS, Jenkins, Terraform
Software Engineer	Data Structures, Algorithms, OOP, Git, SQL, System Design
Data Analyst	Excel, SQL, Python, Tableau, Power BI, Statistics, Pandas
ML Engineer	Python, TensorFlow, PyTorch, Machine Learning, Docker, REST API, SQL

8. SETUP & RUN INSTRUCTIONS

Prerequisites:

- Python 3.10 or higher installed
- pip package manager
- Internet connection (first run only)

Step 1: Clone the Repository

```
git clone https://github.com/ankit-kr6/resume-analyzer  
cd resume-analyzer
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Run the Application

```
streamlit run app.py
```

Step 4: Use the Application

- Browser opens automatically at <http://localhost:8501>
- Upload any PDF resume using the file uploader button
- Select your target job role from the dropdown menu
- View matched skills, missing skills, and match score instantly
- Use sample_resume.pdf included in repo for quick testing

9. CODE EXPLANATION

9.1 analyzer.py – Core Logic

This module contains four key functions that form the backbone of the application:

Function	Purpose
extract_text_from_pdf(pdf_path)	Opens PDF using pdfplumber and extracts all text page by page
load_skills_db()	Reads skills_db.json and returns job role to skills dictionary
extract_skills_from_resume()	Case-insensitive keyword matching to find skills in resume text
get_skill_gap()	Compares resume skills vs required skills, returns matched and missing

9.2 app.py – Streamlit Application

The main application file uses Streamlit to create an interactive web interface. Key components include:

- st.file_uploader() – handles PDF upload from user
- st.selectbox() – provides job role selection dropdown
- st.columns() – creates two-column layout for matched vs missing skills
- st.progress() – displays animated match score progress bar
- st.success() / st.error() – color-coded skill display

9.3 skills_db.json – Skills Database

A JSON file mapping 6 job roles to lists of required skills. This design allows easy extension – new roles and skills can be added without modifying any Python code. The database currently covers 38 unique skills across 6 job roles.

10. SYLLABUS CONCEPTS APPLIED

Concept	How Applied in Project
File Handling & I/O	PDF file reading using pdfplumber, JSON file loading
Functions & Modularity	Separate functions in analyzer.py for each task

Data Structures	Lists for skill arrays, dictionaries for skills_db
String Processing	Case-insensitive skill matching using .lower()
List Comprehensions	Skill extraction and gap detection using comprehensions
JSON Handling	Loading and parsing skills database from JSON file
Web App Development	Full Streamlit web application with interactive UI
Conditional Logic	Score-based feedback messages (if/elif/else)
Module Imports	Importing pdfplumber, streamlit, json, os modules

11. CHALLENGES FACED

PDF Text Extraction Inconsistency

Different PDF formats and generators produce text with varying formatting. Some resumes had text extracted with extra spaces or merged words. Solved by using .lower() for case-insensitive matching and simple keyword search.

Streamlit CLI Executability

Ensuring the app could be launched purely from command line without any GUI setup was a key requirement. Streamlit's 'streamlit run app.py' command satisfies this as it runs a local server and auto-opens the browser.

Skill Matching Accuracy

Simple keyword matching sometimes produced false positives. For example, 'R' (programming language) could match in unrelated words. Mitigated by using full skill names in the database rather than abbreviations.

Temporary File Management

Streamlit's file uploader returns a BytesIO object, not a file path. Had to save the uploaded file temporarily as temp_resume.pdf before passing to pdfplumber for processing.

12. FUTURE SCOPE

- AI-powered semantic skill matching using NLP models (spaCy, BERT) instead of keyword matching
- Integration with LinkedIn API to fetch real-time job skill requirements

- Support for DOCX and TXT resume formats in addition to PDF
- Personalized learning path recommendations based on identified skill gaps
- Resume improvement suggestions with specific course recommendations
- Multi-language support for resumes in Hindi and other regional languages
- Dashboard to track skill development progress over multiple resume uploads
- ATS (Applicant Tracking System) score simulation for job applications

13. CONCLUSION

The Resume Analyzer + Skill Gap Detector successfully achieves its objective of providing students and job seekers with a quick, automated way to understand their skill gaps for target job roles. The application is fully functional, executable via a single command line instruction, and provides clear visual feedback through an intuitive web interface.

This project demonstrates practical application of multiple programming concepts including file handling, string processing, data structures, JSON manipulation, and web application development – all of which are core components of the CSA2001 course syllabus. The modular code structure ensures easy maintenance and extensibility for future enhancements.

Beyond the technical implementation, this project addresses a real-world problem faced by millions of job seekers and provides a foundation that can be scaled into a comprehensive career guidance tool with AI-powered recommendations and real-time job market data.

REFERENCES

1. Streamlit Documentation – docs.streamlit.io
2. pdfplumber Library – github.com/jsvine/pdfplumber
3. Python Official Documentation – docs.python.org
4. GitHub Repository – github.com/ankit-kr6/resume-analyzer
5. JSON in Python – docs.python.org/3/library/json.html
6. Streamlit File Uploader API – docs.streamlit.io/library/api-reference

CSA2001 – Bring Your Own Project (BYOP) | VIT Bhopal University | Ankit Kumar Mandal |
github.com/ankit-kr6/resume-analyzer